Workforce Planning Model

data bytes

UIUC Statistics Datathon 2026

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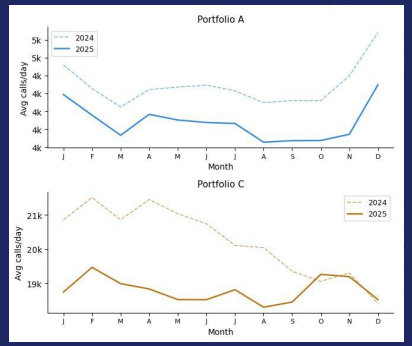
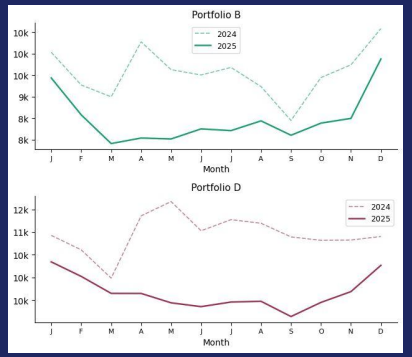
Portfolio A **Portfolio C**



Portfolio B **Portfolio D**



Data Quality Assessment

Finding Missing Values in the Data

Daily Nulls (% of 731 rows)

Portfolio	Call Vol	CCT	Service Level	Abandon Rate
A	3.3%	3.3%	3.4%	4.1%
B	2.7%	4.0%	7.1%	3.4%
C	2.7%	1.9%	2.3%	1.4%
D	1.4%	4.4%	3.0%	—

Interval Nulls (% of rows)

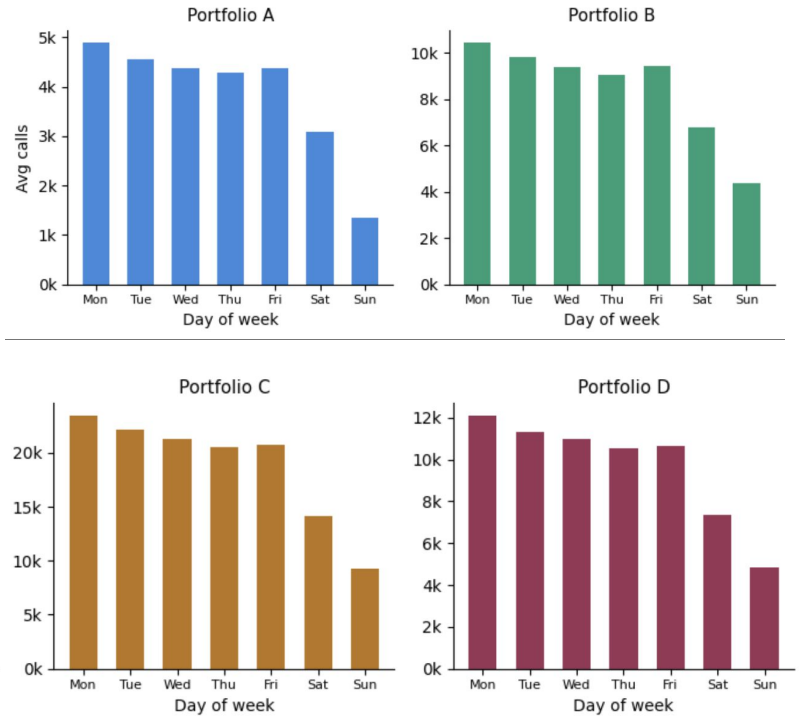
Portfolio	Call Vol	CCT	Abandon Rate	Service Level
A	2.2%	4.1%	4.0%	3.6%
B	2.5%	3.0%	3.1%	3.2%
C	1.8%	1.3%	1.3%	1.3%
D	1.7%	6.5%	6.6%	6.5%

Null Windows Identified:

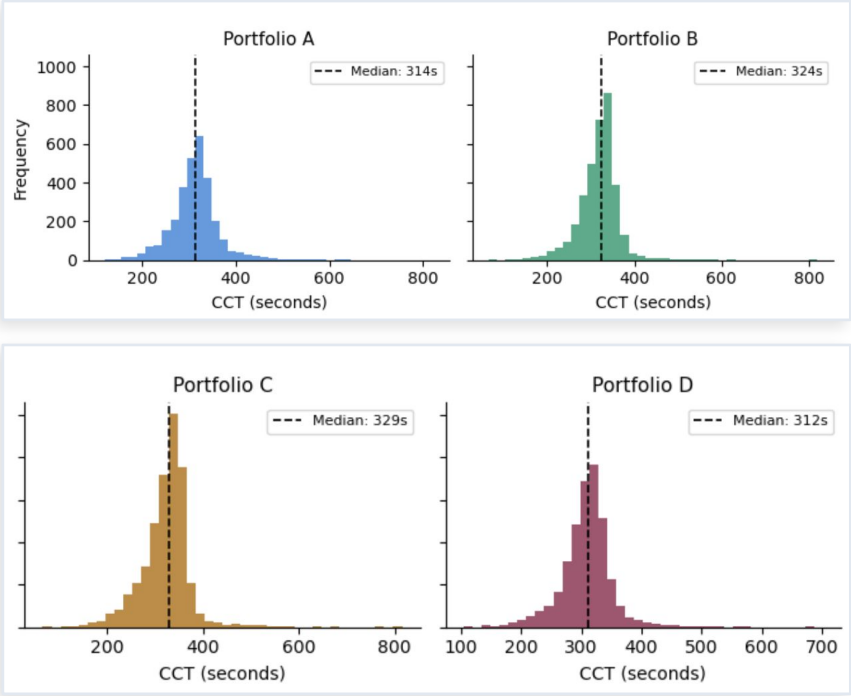
- **Portfolio A:** Sep 5–16 2024 (12 days), Jun 12–18 2025 (7 days) : consecutive window suggests system outage
- **Portfolio B:** Sep 2–11 2024 (10 days), Jun 15–22 2025 (8 days) : similar pattern to A, likely same event
- **Portfolio C:** May 5–11 2024 (7 days), Dec 4 2024 : scattered pattern - shorter outage or batch failure
- **Portfolio D:** Abandon Rate occasionally missing; CCT has 4.4% gap : D has more CCT/ABD dropouts than others

Day-of-Week Patterns & CCT Duration Shape

Day-of-Week Call Volume



CCT Duration Distribution (in seconds)



Forecasting Rationale : Why Statistical Models Win Here

97-99 % of CV fraction variance explained by (DOW × time slot) across portfolios	Trimmed mean is an optimal estimator. GBM would overfit the remaining 1–3% noise.
84-86 / 91 days clean training days per portfolio after partial/incomplete day removal	Smaller data, a simple model works. Bias-variance tradeoff favors low-complexity estimators.
50 - 69 % of interval slots have zero abandon rate	Standard regression fails. Zero-inflated model separates rows with 0 abandon rate from daytime scaling.
69 - 104 secs CCT intraday spread vs ~315s mean 22–33% relative spread	Ratio calibration to daily anchor sufficient, no need for ML to predict CCT.
OLS Regression for intra-month trend , applied only when $R^2>0.40$ and $p<0.10$	Significance gate prevents fitting noise. Without gate, tiny DOW samples yield spurious slopes.

Forecast Pipeline : Methodology

1 Corrupt Day Removal

Remove interval days missing any peak-hour slot (9am–5pm) or
where interval CV sum < 90% of daily total.
Result: 84–86 clean training days per portfolio out of 91 calendar days.

2 Call Volume Shape using Trimmed Mean (10%)

For each (Day of Week, 30-min slot)
Trimmed mean of cv_frac = slot_CV / daily_CV across all clean training days.
Trim removes top and bottom 10% of days. Normalize so each DOW shape sums to 1
Slot forecast = daily_anchor × shape_weight.

3 Newsvendor Uplift - 67th percentile of residuals

Leave-one-out DOW residuals from same-DOW mean. Uplift = q_{67} / mean_CV
Results: A= 3.0% B= 5.8% C= 4.4% D= 6.0%.
Applied to business hours (8am–8pm) only and half-uplift on shoulder hours
(6 – 8am, 8 – 10pm).

4 Ordinary Least Square Regression Intra Month - Significance Gate

Fit OLS of daily Call Volve vs occurrence-within-month per DOW.
Only apply if R^2>0.40 AND p<0.10 and |slope/mean|>1.5%.
Without this gate, tiny DOW samples (≤5 rows) would generate spurious trends.

5 Spring Seasonal Ratio for August

Compute Aug/Spring DOW-level ratio using 2024 and 2025 data.
Applied to shape profiles to account for temporal shift.
Range: 0.900–1.109 across portfolios. Scaling handled by daily anchor.

6 CCT & ABD Calibration

CCT: (DOW, slot) ratio profile × daily CCT anchor. Clipping 60 - 1,200s.
ABD: Zero-inflated : overnight slots (before 7am, after 10pm) forced to 0
Daytime proportional to daily ABD anchor.

Newsvendor Uplift & Model Validation

Newsvendor Critical Ratio Formula

Cost of under-forecast (α=2): Under-staffing causes SL failure, customer abandonment
Cost of over-forecast (β=1): Excess idle capacity, less costly
Critical Ratio = α/(α+β) = 2/3 : Find 67th percentile of residuals
Residuals computed via leave-one-out cross-validation by DOW
Using all available training months 2024 and 2025

Computed Uplifts

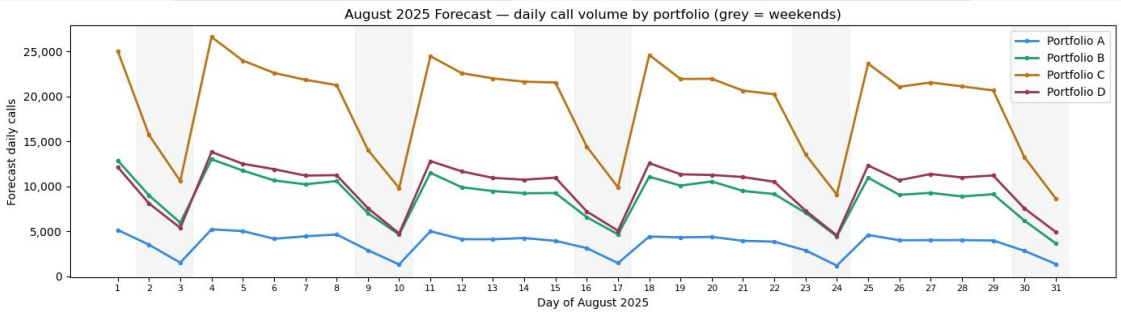
A	n=170 q_{67}=123 mean=4,118	**3.0 %**
B	n=174 q_{67}=522 mean=8,984	**5.8 %**
C	n=175 q_{67}=881 mean=19,802	**4.4 %**
D	n=182 q_{67}=621 mean=10,420	**6.0 %**

Forecast vs Anchor : Expected Over-Forecast = Uplift %

Portfolio A	Portfolio B	Portfolio C	
Daily Expected CV	Daily Expected CV	Daily Expected CV	Daily Expected CV
2.9%	**5.3%**	**4.1%**	**5.4%**

August 2025 Forecast : Final Results

Portfolio A	Portfolio B	Portfolio C	Portfolio D
Aug Total Calls	Aug Total Calls	Aug Total Calls	Aug Total Calls
113,797	**275,474**	**590,498**	**305,872**



Month	Day	Interval	CV_A	ABD_A	CCT_A	CV_B	ABD_B	CCT_B	CV_C	ABD_C	CCT_C	CV_D	ABD_D	CCT_D
August	1	0:00	6.46	0	371.27	40.31	0	345.25	82.96	0	372.98	44.92	0	339.65
August	1	0:30	4.87	0	357.06	30.29	0	318.06	56.22	0	335.09	31.59	0	306.32
August	1	1:00	3.49	0	330.26	13.27	0	465.68	43.22	0	343.3	22.71	0	382.4

Code Quality & Structure

Innovation	GBM rejected because it would fit noise more Replaced 6% fixed uplift with newsvendor formula, zero inflated model used instead of standard regression
Feasibility	Runs across all 4 portfolios in seconds Scales linearly with portfolio count with no external dependencies required
Flexibility	Each component is a standalone function, no code changes needed to recalibrate Daily anchor has a two-tier fallback system
Trust & Explainability	Every forecast slot traces back to: daily anchor × DOW shape × uplift scalar: fully auditable Validation checks for all outputs including null counts

Thank You!